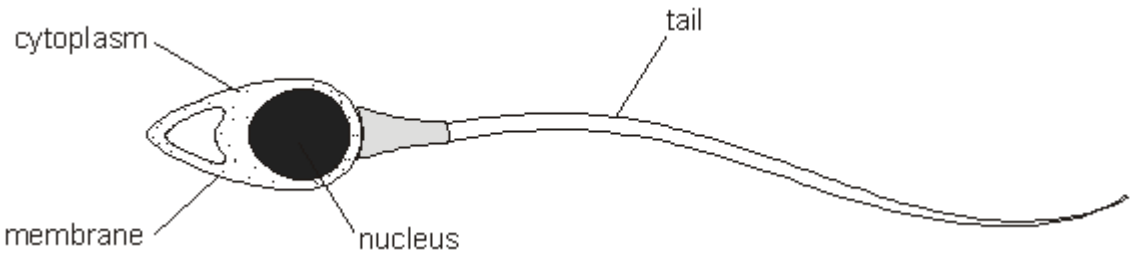
Q1. (a) The diagram shows a sperm cell. Sperm cells are adapted for fertilisation.



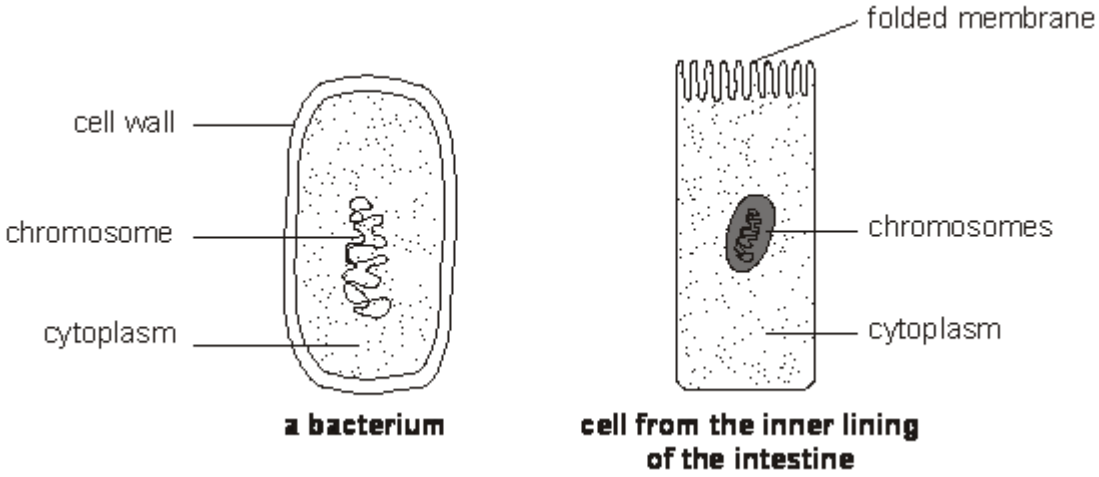
Sperm cells use their tails to swim towards an ovum (egg).
Give **one** other way the sperm cell is adapted for fertilisation.

.....

.....

1 mark

(b) The diagrams below show two other cells.



a bacterium

cell from the inner lining of the intestine

not to scale

(i) Look at the diagrams above.

What is the difference between the location of the genetic material in the bacterium and in the cell from the lining of the intestine?

.....

.....

.....

1 mark

(ii) What is the function of the genetic material in a cell?

.....

.....

1 mark

(c) Cells in the lining of the intestine are adapted to absorb digested food.

How does the folded membrane of these cells enable them to absorb the maximum amount of digested food?

.....

.....

1 mark

(d) A group of cells in the lining of the intestine is a tissue.

Why is a number of sperm cells **not** a tissue?

.....

.....

1 mark

maximum 5 marks

Mark schemes

Q1.

(a) any **one** from

- it is streamlined
  *do **not** accept 'it is slimline'*
- it has a pointed head
  'its shape' is insufficient
- it does not have much cytoplasm
  accept 'there are enzymes in the head'
  'it has a nucleus' is insufficient as it is not a specific adaptation of a sperm cell

1 (L7)

(b) (i) any **one** from

- in the cytoplasm of the bacterium
- in a nucleus **or** within a membrane in the cell from the intestine
  *accept 'the bacterium does not have a nucleus **or** nuclear membrane'*
  accept 'in the bacterium it is not enclosed'

1 (L7)

(ii) any **one** from

- it determines the characteristics of the cell
- it transfers information
  accept 'gives the cell instructions'
  accept 'it tells the cell what to do'
  accept 'to control the cell'
  accept 'allows the cell to reproduce'

1 (L7)

(c) any **one** from

- it has a large surface area
- the folds increase the surface

1 (L7)

(d) they are independent cells **or** they do not work together **or** do not work in a group
*accept 'each sperm cell is individual **or** works individually'*
'the cells are not joined together' is insufficient

1 (L7)

[5]